Climbing the Ladder: Automating Data Pipelines and Reproducing Baselines in Low-Resource Settings

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Abstract

The massively multilingual translation systems that now include Sicilian learned it from noisy, non-standardised web text. To obtain better translation quality, we redevelop the small neural machine translation (NMT) system that Wdowiak (2022) customized for Sicilian. We rebuild the parallel-corpus pipeline with sentence-embedding extraction and alignment (PyMuPDF and LaBSE (Feng et al., 2022)), assemble a reproducible, deduplicated Sicilian–English dataset with a held-out *literary* test set, and port the paper's tokenization and desinence-biasing recipe to a Perl-free stack. We evaluate both from-scratch Sockeye baselines and the pretrained NLLB-200 model (NLLB Team, 2022) adapted with LoRA (Hu et al., 2021). On our harder held-out test, Sicilian→English climbs from a 5.27 BLEU floor to 31.43 with a bidirectionally fine-tuned NLLB-1.3B—above Wdowiak's published baseline (29.1)—while the reverse direction nearly doubles (9.89→18.73). We further include **Italian** as a closely-related, higher-resource bridge language (`ita_Latn`): NLLB supports it↔scn natively, and we evaluate and fine-tune those directions on a frozen WikiMatrix it–scn split. The full pipeline, dataset assembly and evaluation are reproducible.

1 Introduction

Languages without an army and navy often lack a standard orthography. Without specific spelling rules and conventions, written texts in these languages often exhibit great variability (sometimes even within a local dialect), which inhibits the ability of conventional metrics to assess model performance (Aepli et al., 2023). But even languages like Sicilian – where published dictionaries have presented a consistent orthography for several hundred years (Pasqualino, 1785; Mortillaro, 1876; Nicotra, 1883; Traina, 1868) – can exhibit extreme orthographic diversity which (when incorporated into a training dataset) prevents a model from learning how to generate consistent text.

During the 13$^{\text{th}}$ century, the Sicilian School of Poets at the court of Emperor Frederick II created the first literary standard in Italy. Their poetry inspired Dante to write in his native Florentine, which later became today's Italian language. Following in their footsteps, poets like Antonio Veneziano (1543-1593), Giovanni Meli (1740-1815), Domenico Tempio (1750-1821), Ignazio Buttitta (1899-1997) and Salvatore Camilleri (1921-2021) continued the Sicilian literary tradition, writing in a fairly consistent manner. But writing is different from speaking, so Giuseppe Pitrè (1841-1916) developed a popular tradition which recorded the common people in their own words and in the sound of their own words.

One can hear echoes of the popular tradition in the daily speech of millions of people in Sicily, Calabria and Puglia, as well as in diaspora communities abroad. And one can find echoes of the literary tradition in books, journals and pages of the internet. Nonetheless, the publicly available Sicilian language text often exhibits extreme orthographic diversity which once prevented practitioners from developing datasets of Sicilian-language parallel text and once caused Sicilian to be absent from every major translation service.

Today, Sicilian is included in massively multilingual models such as NLLB-200 (NLLB Team, 2022), but the coverage is *low-quality*: the Sicilian training text was mined from non-standardised web sources, so the model has, in effect, learned a non-standard orthography (Section 4). The barrier for Sicilian has thus shifted from *absence* to *quality*.

High-quality material exists, especially in the bilingual literary journal *Arba Sicula* (published since 1979), which Wdowiak (2022) used to create the first Sicilian neural machine translation (NMT) system. His work demonstrated that it's possible to develop a fairly good translation model in low-resource settings, but it was a labor-intensive task. He aligned sentences by hand using a recipe tailored to the low-resource, morphologically rich setting.

This work rebuilds that effort with maintained, mostly CPU-only tooling and seeks to obtain better quality with modern pretrained models.

Contributions.

- An automated extraction and alignment pipeline for the *Arba Sicula* PDFs (PyMuPDF and LaBSE (Feng et al., 2022)), replacing the legacy pdflatex+hunalign flow, with thresholds tuned against Wdowiak's hand-aligned gold.
- A reproducible, deduplicated, provenance-tracked dataset with a held-out literary test set (not FLORES), and a BLEU/chrF evaluation harness.
- The paper's "lever B" recipe (Napizia tokenizer + desinence-biased subwords) ported to a Perl-free pipeline, and a linguistic-feature lever (lemma source factors).
- NLLB-200 zero-shot and LoRA fine-tuning that beats the published baseline on our harder test set, with a bidirectional adapter that also rescues the reverse direction.

2 Background

NMT models translation as a conditional language model, in which an encoder reads the source $x = (x_1, \ldots, x_n)$ and a decoder factorises the target from left to right, $p_\theta(y \mid x) = \prod_t p_\theta(y_t \mid y_{<t}, x)$, trained by maximising the log-likelihood of a parallel corpus and decoded with beam search. The dominant parameterisation is the **Transformer** (Vaswani et al., 2017), which replaces recurrence with scaled dot-product attention, $\mathrm{Attention}(Q, K, V) = \mathrm{softmax}(QK^\top/\sqrt{d_k})V$, run in parallel heads.

With little data the default Transformer overfits; the low-resource recipe of Sennrich and Zhang (2019) prescribes a smaller, heavily regularised model with high dropout (Srivastava et al., 2014) and small subword vocabularies. **Subword segmentation** via byte-pair encoding (Sennrich et al., 2016) gives an open vocabulary and shares statistics across word forms—essential for a morphologically rich language. Further gains come from **linguistic input features** (Sennrich and Haddow, 2016), **back-translation** of monolingual data (Sennrich et al., 2015), and **multilingual** training with a target-language token (Johnson et al., 2016) that enables bridging through related high-resource languages (Fan et al., 2020; Arivazhagan et al., 2019). Low-resource MT remains hard for exactly these reasons (Koehn and Knowles, 2017).

3 Data Pipeline

3.1 Extraction from PDFs

We extract clean text from the Arba Sicula issue PDFs with PyMuPDF (no pdflatex round-trip). Each page is assigned a language from its share of Sicilian stop-words and diacritics; the journal sets Sicilian and English on facing pages, giving a natural even/odd prior, so we form facing candidate pairs. Pages are segmented into sentences with `pysbd` (per language), filtering "furniture" (all-caps headers, page numbers, captions). The procedure recovers parallel pages the manual selection had skipped.

3.2 Sentence alignment

We embed every sentence with LaBSE (Feng et al., 2022), a language-agnostic dual encoder, and build the cosine-similarity matrix $S_{ij} = \langle u_i, v_j \rangle / (\|u_i\| \|v_j\|)$. A facing pair is confirmed only if its mean best-match similarity exceeds τ_{page}; sentences are then aligned by a monotonic dynamic program over S

Table 1: Unified dataset.

split	pairs	source
train	27,386	Arba Sicula + NLLB
valid	1,000	Arba Sicula
test	1,000	Arba Sicula (literary)
it–scn (aux)	11,389	WikiMatrix

allowing 1–1, 1–2 and 2–1 moves with a null penalty, in the spirit of Vecalign (Thompson and Koehn, 2019) but without a bilingual dictionary or hunalign. The thresholds $(\tau_{\text{page}}, \tau_{\text{sent}}) = (0.50, 0.40)$ were tuned against Wdowiak's hand-aligned gold (issues AS41–42); the pipeline recovers 14,245 sentence pairs from 44 issues, comparable to his manual alignment. Crucially, the *unaligned* English is retained as an in-domain monolingual pool for back-translation.

3.3 Additional sources

We add the author's curated "Good-Sicilian-in-NLLB" set—LASER3-mined sentences (Heffernan et al., 2022; NLLB Team, 2022) rescored and filtered for quality and Corsican contamination—a small curated "Good-Sicilian-from-WikiMatrix" set, and the WikiMatrix it–scn subset (Schwenk et al., 2021) as an auxiliary Italian bridge.

3.4 Unified dataset

Sources are cleaned, deduplicated across one another, and split deterministically. Then a 1,000/1,000 valid/test split is *frozen* once, held out from Arba Sicula (which provides the high-quality literary standard that FLORES does not), so that every later result stays comparable (Table 1).

4 Standardization

Although Sicilian does not have a single official orthography, a common literary standard has existed for several hundred years and provides a rough consistency, which we can refine because the majority of the Sicilian sentences in the NLLB dataset were written in that standard.

Nonetheless, approximately one-third of the Sicilian sentences that the NLLB team collected were written in local dialect. Some of the sentences were written in Corsican. And very few of the Sicilian sentences in the NLLB dataset were paired with a valid translation. We suspect that these data quality issues were caused by seed data which is inconsistent with the literary standard.

4.1 Why Sicilian lives in the Romance neighbourhood

A pretrained multilingual encoder needs no explicit Sicilian support to *read* Sicilian because the multilingual representations that they learn is not a shared interlingua, where sentences from different languages get mapped into an abstract, universal representation of meaning. Instead they organise their parameter space around surface-form cues (Verma et al., 2026).

The surface form of literary Sicilian is close to Italian, so reading it works well. On the multi-parallel FLORES-200 devtest (NLLB Team, 2022), LaBSE (Feng et al., 2022) retrieves the correct English translation of a Sicilian sentence **99%** of the time (precision@1, Sicilian→English), one point behind *supported* Italian (100%); and, discounting English—LaBSE's pivot, to which everything is close—each Sicilian sentence's nearest neighbour is overwhelmingly Italian (18%, twice the next Romance language). In lexicon and spelling Sicilian is close enough to Italian that the model treats it almost as a variety of Italian.

The seed-data failure, however, is in the *opposite* direction – retrieving the Sicilian translation of an English sentence. Romance languages share a lot of vocabulary and grammar, so when a model retrieves a Sicilian, Italian or Corsican translation of an English sentence, the translation that it selects will reflect the surface-form cues in its training data. Sometimes the NLLB seed data's surface form reflects the literary language in Sicilian dictionaries, but other times it reflects local pronunciations, which (we suppose) is what caused their model to frequently retrieve Corsican-language sentences when assembling the NLLB's Sicilian corpus.

Since multilingual models encode script and surface form over deeper linguistic structure (Verma et al., 2026), the two directions diverge for the same reason: an Italianate surface form retrieves the right English, while Corsican-looking seed data makes the model generate the wrong Sicilian. The proximity is thus double-edged: it is exactly what lets our own LaBSE-based aligner (Section 3.2) operate on an officially unsupported language, but it offers no protection against near-language contamination when a corpus is assembled by transfer alone.

Crucially, alignability tracks *standardness*. Scoring the LaBSE cosine of Wdowiak's hand-aligned gold pairs per text, standard literary prose (Cipolla's *Don Chisciotti*) aligns at cosine **0.76**, with 91% of its pairs clearing our production threshold, whereas dialect poetry aligns at only **0.64** (80% recovered). The non-standard material is thus both the larger source of noise in the collected data and the harder to recover, so curating toward the literary standard improves quality on both counts. Rolling short verses into longer lines recovers much of the poetry gap (mean cosine $0.65 \rightarrow 0.76$, recovery $84\% \rightarrow 99\%$ when merging up to three verses), a cheap corrective for the poetic register.

For a translation model, we need sentences written in the Sicilian literary language and we need translations of those sentences. So we back-translated all of the Sicilian sentences into English and Italian, then we scored the resulting pair on the task of translation into Sicilian. because a sentence written in standard Sicilian will score better than a sentence written in local dialect, forward-scoring the back-translations allows us to identify the sentences written in standard Sicilian. This method yielded 549,486 back-translations to simulate English-to-Sicilian translation and 689,840 to simulate Italian-to-Sicilian translation.

5 Tokenization and Subword Segmentation

Because the majority of the Sicilian sentences in the NLLB dataset were written in a common literary standard, we can refine them for consistency. We first **normalise** text toward the standard established by Cipolla and augmented by Wdowiak for MT clarity: strict **L** on articles (*lu*/*la*/*li*), **h** on *aviri* (*haiu, havi, hannu*), **CI** for çi (*çiuri*→*ciuri*), and **R** retained before a clitic (*farlu*). Our `normalize_scn.py` applies the documented spelling rules (e.g. *cchiù*→*chiù*, *io*/*iu*/*eu*→*jo*, *non*→*nun*) at a "std" level, and additionally uncontracts and ASCII-folds at a "full" level. Because NLLB's own tokenizer was trained on un-normalised web Sicilian, whether this normalisation *helps* the pretrained model is an empirical question. We ablate it (scn→en, identical English references): **raw 30.60, std 30.98, full 30.98** BLEU (chrF 56.31 / 56.63 / 56.50). The light "std" standardisation gives a small consistent gain (and the best chrF), while the aggressive "full" preprocessing does *not* improve over it—consistent with the pretrained tokenizer expecting un-normalised input. Orthographic normalisation is thus a *minor* lever for NLLB ($\sim$+0.4 BLEU); we adopt std by default, and conclude that the larger residual relative to the state of the art is *data curation*, not spelling.

We re-implement the Napizia tokenizer in pure Python (verified identical to the original Perl): it ASCII-folds accents and *uncontracts* Sicilian (e.g. ô→*a lu*, dû→*di lu*), cutting sparsity. "Lever B" then biases the joint scn–en BPE (Sennrich et al., 2016) toward the textbook *desinences* catalogued in the resources Wdowiak assembled to standardise Sicilian—Dieli's *Sicilian Vocabulary*, Cipolla's *Mparamu lu sicilianu* (Cipolla, 2013), Bonner's grammar (Bonner, 2001), and the *Chiù dâ Palora* dictionary—so morphological endings survive as units (Wdowiak's example: *Carinisi are dogs!* segments to `car@@ in@@ isi are dogs !`). This cut the vocabulary from 14,000 to 2,000 units and produced his largest BLEU gains. BPE-dropout (Provilkov et al., 2020) adds subword regularisation by sampling segmentations during training.

Table 2: Our models, our test set (scn→en).

model	BLEU	chrF
floor (copy source)	5.27	25.40
Sockeye baseline	5.54	28.28
+ lever B (tok. + desinences)	7.24	29.52
+ more data	9.79	33.82
+ lever D (lemma source factors)	10.85	35.22
NLLB-200 1.3B, zero-shot (raw)	29.02	55.23
NLLB-200 1.3B, LoRA fine-tuned, scn→en only	31.16	56.79
NLLB-200 1.3B, LoRA *bidirectional*	31.43	56.94
+ back-translation (7.5k synthetic)	**31.60**	**57.21**

6 Models

The from-scratch baseline is a Sockeye-3 (Hieber et al., 2017) small Transformer (3 layers, model size 256, 4 heads, feed-forward 1024, weight tying, dropout 0.4, joint BPE ≈4k; ∼6.6M parameters), trained on CPU. On top of it we stack lever B (tokenization + desinences), more parallel text, and "lever D" linguistic input features (Sennrich and Haddow, 2016): a lemma source factor summed into the source embedding, $e_j = E_{\text{word}}(w_j) + E_{\text{lemma}}(\ell_j)$, whose small (∼900) vocabulary generalises across inflected forms.

The pretrained model is NLLB-200 (NLLB Team, 2022), a massively multilingual encoder–decoder that supports Sicilian (`scn_Latn`) and is steered by forcing the first decoder token. We adapt the 1.3B variant with LoRA (Hu et al., 2021): freezing W_0 and learning a low-rank update $W = W_0 + \frac{\alpha}{r}BA$ (r=32, α=64) on the attention projections, with the base in fp16 and the adapters in fp32. Training *both* directions at once (scn→en and en→scn) yields a single adapter that serves both and rescues the weak reverse direction.

7 Evaluation

We score with sacreBLEU (Post, 2018)—BLEU (Papineni et al., 2002) (`tok:13a`) and chrF (Popović, 2015)—on the frozen literary test set rather than the NLLB/FLORES set: the FLORES Sicilian uses the post-2017 orthography and a different (non-literary) domain, so it is not representative of the Arba Sicula material. The Sockeye rows are scored in tokenized space (the same space as the upstream system; the raw-text floor is 5.27 BLEU), while the NLLB rows are scored on raw text.

8 Results

On the frozen held-out test set (1,000 literary pairs), Sicilian→English results are reported in Table 2. The Sockeye levers stack from 5.54 to 10.85 BLEU on CPU alone; the pretrained NLLB-1.3B wins decisively even zero-shot (29.02), and LoRA fine-tuning lifts it to 31.43—above Wdowiak's published baseline (29.1) on our harder test.

In the reverse direction (en→scn), the same bidirectional adapter lifts NLLB-1.3B from 9.89 (zero-shot) to 18.73 BLEU / 49.96 chrF, nearly doubling the weak direction at no cost to scn→en and giving a usable two-way model. This is still below the published en→scn baseline (25.1); back-translation is the next lever.

Italian directions (it↔scn). Italian is a closely related, much higher-resource Romance language, and NLLB supports it↔scn natively. We hold out a frozen 1,000-pair it–scn test split from the WikiMatrix data, evaluate NLLB-1.3B zero-shot, then fine-tune *all four* directions jointly (scn↔en and it↔scn) into a single multilingual adapter, a bridge in the sense of Johnson et al. (2016) and Fan et al. (2020)

Table 3: Trilingual model: all directions, zero-shot vs. the multilingual fine-tune (BLEU / chrF, frozen test sets).

direction	zero-shot	multilingual FT
scn→en	29.02 / 55.23	30.75 / 56.57
en→scn	9.89 / 41.12	17.49 / 48.78
it→scn	17.61 / 44.77	**26.97 / 51.69**
scn→it	42.20 / 59.04	**43.47 / 59.79**

(Table 3). Italian is by far the easiest pair—scn→it reaches **43.5** BLEU and it→scn climbs $17.6 \rightarrow$ **27.0**—as expected for a close, high-resource neighbour on in-domain WikiMatrix text. The multilingual adapter costs a little on scn↔en ($30.8/17.5$ vs the dedicated $31.4/18.7$), so one can ship either the trilingual all-rounder or the specialised pair.

Relation to the state of the art. We do *not* beat the state of the art. Wdowiak's **reverse-training** system reports, on his own in-domain hand-selected test set, En→Sc/Sc→En BLEU of 45.1/48.6 and It→Sc/Sc→It of 61.4/62.9—well ahead of our 18.7/31.6 and 27.0/43.5 on every direction (his earlier marks were 25.1/29.1 baseline, 35.0/36.8 augmented). The numbers are *not* a head-to-head (different test sets), but the gap is too large to be explained by that alone: on Sc→en we only clear his *baseline* (31.6 vs 29.1), on a harder test, and trail elsewhere. Crucially, his reverse-training builds a *custom pre-trained model* from tens of millions of pairs (forward-translation, then back-translation, in three stages) before fine-tuning on *Arba Sicula*—a pipeline infeasible to reproduce on a single GPU. NLLB-200, however, already provides that massive multilingual pretraining: our fine-tune is the analogue of his stage-3. We applied the feasible parts of his recipe on top of NLLB and each adds little: the multilingual Italian bridge, orthographic normalisation ($+0.4$ BLEU), and back-translation ($+0.17$, capped by our 7.6k in-domain monolingual pool versus his ~550k). The *method* gap is therefore largely closed; the residual is **data**—his hand-curated, standardised corpus, far more back-translation, and an in-domain test set—which argues for combining his corpus with our pipeline rather than competing.

9 Conclusion and Future Work

We presented a reproducible, Perl-free path from Arba Sicula PDFs to a Sicilian–English NMT system: an automated extraction-and-alignment pipeline, the paper's tokenization and desinence recipe with an added linguistic-feature lever, and NLLB-200 adapted with LoRA. On our harder literary test set the best model reaches 31.6 BLEU (scn→en), above the published baseline but well below Wdowiak's reverse-training state of the art. Having applied the feasible parts of his recipe on top of NLLB—**multilingual** Italian bridging, orthographic **normalisation**, and **back-translation**—each for only a small gain, we conclude the residual is *data*: his hand-curated corpus, far more back-translation, and an in-domain test set. This is a concrete case for combining his corpus with our pipeline. We also release **serving** (a local web app and a Telegram bot, a FastAPI service over the NLLB adapter). All code, the data pipeline and the evaluation harness are released for reproducibility.

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